



**BHASKARACHARYA NATIONAL INSTITUTE FOR SPACE
APPLICATIONS AND GEO-INFORMATICS**
WEEKLY PROGRESS REPORT (06/07/2026 – 12/07/2026)

WEEK 8

PROJECT NAME	ROAD CONDITION DETECTION
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PROJECT DESCRIPTION :

**AUTOMATIC DETECTION AND
SEGMENTATION OF ROAD DAMAGES**

GROUP MEMBER :

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GROUP ID :

2026G263

GROUP GUIDE :

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06/07/2026	Started preparing the project report by documenting the system architecture and overall workflow of the RoadSense application.
07/07/2026	Added important source code (CD) snippets and explanations for the report appendix. Documented key functions used in the inference pipeline.
08/07/2026	Prepared the weekly progress report, updated completed work, and organized screenshots and output results for documentation.
09/07/2026	Reviewed and refined the project documentation. Updated module descriptions, flowchart explanations, and implementation details.
10/07/2026	Continued report preparation by adding methodology, testing results, references, and formatting the report according to internship requirements.
11/07/2026	Holiday
12/07/2026	Holiday (Sunday)

REFERENCE:

1. Jocher, G., et al., *Ultralytics YOLO Documentation*. Available: <https://docs.ultralytics.com/>
2. Long, J., Shelhamer, E., & Darrell, T., "Fully Convolutional Networks for Semantic Segmentation," *CVPR*, 2015.
3. Ronneberger, O., Fischer, P., & Brox, T., "U-Net: Convolutional Networks for Biomedical Image Segmentation," *MICCAI*, 2015.
4. The Python Software Foundation, *Python Documentation*. Available: <https://docs.python.org/3/>

SCREEN SHOTS:

RoadSense
AI Road Inspection

Dashboard

Detections

Analytics

History

OVERALL HEALTH

80

GOOD

ROAD CONDITION

Detections

30

RS

RoadSense User

ADMIN@ROADSENSE.COM

New Session

Deploy

Detection Log

Select session

Current Session

Filter by class

Alligator Cracking > Crack > Guard Rail > Line Marking > Pothole > Ravelling > Rutting > Traffic Sign > Tree > >

Filter by priority

URGENT > SCHEDULE > MONITOR > ASSET > >

10 of 10 detections shown

Sort by column:

Area (m²)

☐ Ascending order

ID	Class	Priority	Area (sq°)	Area (m²)	H-Length (m)	V-Length (m)	Frame	Time (s)	Section	Distance (m)
0	Ravelling	MONITOR		1327	0.0015	None		22	0.74 KM 0.01	9.7
0	Traffic Sign	ASSET		1109	None	None		2	0.07 KM 0.00	0.9
0	Tree	ASSET		24160	None	None		2	0.07 KM 0.00	0.9
1	Tree	ASSET		9754	None	None		2	0.07 KM 0.00	0.9
2	Tree	ASSET		7518	None	None		2	0.07 KM 0.00	0.9
3	Tree	ASSET		10766	None	None		2	0.07 KM 0.00	0.9
4	Tree	ASSET		39703	None	None		4	0.13 KM 0.00	1.8
5	Tree	ASSET		90971	None	None		6	0.2 KM 0.00	2.6
6	Tree	ASSET		28921	None	None		8	0.27 KM 0.00	3.5
7	Tree	ASSET		5777	None	None		10	0.33 KM 0.01	4.4

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